Presentation Plan: From Random Walks to Sequential Testing

Version 4 — Revised After Reading All Papers

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About This Document

Target audience. People comfortable with algebra and basic probability (e.g. a 12th-grade math background). No calculus is required to follow the main ideas, though some calculus notation (integrals) appears in later acts for completeness — always with plain-language translations.

Format. Fourteen interactive acts (Acts 0–13) — deliberately small steps. Each act has a simulation, an intuitive explanation, and a mathematical section. The math sections always define every symbol before using it. Every formula is accompanied by a plain-language translation immediately below it.

Design philosophy. Better to over-explain than under-explain. When in doubt, slow down.

What changed from v3. After reading all seven reference papers in detail, several significant corrections were made. The most important:

- Eppo (Schmit & Miller, 2024) does **not** use the mSPRT framework of Johari et al. (2017). They use **confidence sequences** from Howard et al. (2021), specifically the Normal mixture boundary.
- GAVI does **not** appear in the Eppo paper. The tuning parameter is ν (related to expected sample size), not τ.
- Robbins (1970) is the bridge between Wald's SPRT and modern methods — he introduced the mixture approach and confidence sequences. He now receives his own subsection.
- CUPED details are corrected to match Deng et al. (2013) exactly.
- Wald (1945) details are enriched with his actual ASN formulas and savings results (47–63%).
- The narrative arc is corrected: SPRT → Robbins' mixtures → Johari's mSPRT → Howard's confidence sequences → Eppo's implementation.

1 Act 0 — Why Should You Care? The Peeking Problem

1.1 The Motivation

Intuitive Explanation

You work at a company running an A/B test. Half your users see the old website, half see a new design. Every day, you check the dashboard: "Is the new design better?" After a week, the dashboard shows $p = 0.03$ — that looks significant! You call it: the new design wins.

But here is the dirty secret of traditional statistics: **the more often you check, the more likely you are to see a false positive.** A test designed to have a 5% false positive rate can easily reach 20–30% if you peek at it repeatedly. One in four or five "winning" experiments might actually be noise.

This is not a theoretical concern. It is the single most common statistical mistake in online experimentation.

1.2 The Simulation

Simulation

Two groups of observations (control and treatment) are generated from *identical* distributions — there is no real effect. A standard t-test is computed after every new observation arrives.

A counter shows: "Number of times $p < 0.05$ so far." The user watches the p-value bounce around. Every time it dips below 0.05, it lights up red — a false positive.

Users can control:

- How many total observations to simulate
- How many independent experiments to run in parallel

A bar chart accumulates: "Fraction of experiments that showed $p < 0.05$ at *some* point during peeking." For 1,000 simulated experiments with regular peeking, this fraction climbs well above 5% — often to 20–30%.

A second mode checks only once, at the very end. That fraction stays near 5%.

1.3 Intuitive Explanation

Intuitive Explanation

Traditional statistical tests promise: "If there is no real effect, I will falsely cry wolf at most 5% of the time." But this promise comes with fine print: **you may only look at the result once, at a pre-determined time.**

Every time you peek, you get another chance for randomness to fool you. It is like rolling a die repeatedly — the more rolls, the more likely you are to get a six eventually, even though any single roll has only a 1/6 chance.

The rest of this presentation builds a test that does not have this problem: a test where you can peek as often as you like and the false positive guarantee still holds. That promise has a name: **anytime-valid inference**. Getting there requires some beautiful mathematics, which we will build up one step at a time.

Key Takeaway

The peeking problem: Checking a traditional test repeatedly inflates the false positive rate far beyond its nominal level. We need a fundamentally different kind of test — one that remains valid no matter when or how often we look.

2 Act 1 — The Random Walk

2.1 The Simulation

Simulation

A vertical number line appears on screen. A dot starts at position 0. Each step, a coin is flipped. Heads: the dot moves up one step. Tails: down one step. The path is drawn as a graph (position vs. step number).
User controls: speed, number of steps, probability of going up (initially fixed at 50/50; later a slider), number of simultaneous paths.

2.2 Intuitive Explanation

Intuitive Explanation

Imagine you are standing in the middle of a long hallway. Flip a coin. Heads — step forward. Tails — step back. Repeat hundreds of times.
Where will you end up? We don't know exactly, but we can say something precise about the *range* of likely outcomes. Most of the time you won't wander far. Occasionally, a long streak of heads or tails carries you far in one direction. The longer you walk, the wider the range of possible positions — but the centre of that range stays at zero.
When we run many people simultaneously, they fan out over time — like ink dropped into water. This spreading has a precise mathematical shape.

2.3 Mathematical Formulation

Setting up the notation

Let X_i represent the outcome of the ith coin flip:

$$X_i = \begin{cases} +1 & \text{if heads (step forward),} \\ -1 & \text{if tails (step backward).} \end{cases}$$

The probability of heads is written p. For a fair coin, $p = 0.5$.

Position after n steps

$$\boxed{S_n = \sum_{i=1}^{n} X_i = X_1 + X_2 + \cdots + X_n}$$

Read as: "S sub n equals the sum of all X_i, starting from $i = 1$ up to $i = n$."

Worked example. First 5 flips: H, T, H, H, T, so $X_1 = +1$, $X_2 = -1$, $X_3 = +1$, $X_4 = +1$, $X_5 = -1$.

$$S_1 = 1, \quad S_2 = 0, \quad S_3 = 1, \quad S_4 = 2, \quad S_5 = 1.$$

The average outcome (expected value)

When $p = 0.5$, each flip has expected value zero:

$$\mathbb{E}[X_i] = \underbrace{0.5}_{P(\text{heads})} \times (+1) + \underbrace{0.5}_{P(\text{tails})} \times (-1) = 0.$$

Because each step averages to zero:

$$\boxed{\mathbb{E}[S_n] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \cdots + \mathbb{E}[X_n] = 0}$$

On average, across many trials, you end up back where you started.

How spread out the outcomes are — variance & standard deviation

Variance measures the average squared distance from the mean. For a single fair coin flip:

$$\mathrm{Var}(X_i) = \mathbb{E}\Big[(X_i - 0)^2\Big] = 0.5 \times (+1)^2 + 0.5 \times (-1)^2 = 1.$$

Because flips are independent, the variance of the sum is the sum of the variances:

$$\mathrm{Var}(S_n) = \mathrm{Var}(X_1) + \mathrm{Var}(X_2) + \cdots + \mathrm{Var}(X_n) = \underbrace{1 + 1 + \cdots + 1}_{n \text{ terms}} = n.$$

The **standard deviation** is the square root of the variance — same units as the position:

$$\boxed{\mathrm{SD}(S_n) = \sqrt{\mathrm{Var}(S_n)} = \sqrt{n}}$$

After 100 steps, the typical distance from zero is $\sqrt{100} = 10$. After 10,000 steps, it is $\sqrt{10{,}000} = 100$. The spread grows with the *square root* of the number of steps, not the number itself.

The simulation shows the $\pm\sqrt{n}$ envelope as two curved lines.

What if the coin is not fair?

When $p \neq 0.5$, each step has a non-zero average:

$$\mathbb{E}[X_i] = p \times (+1) + (1 - p) \times (-1) = 2p - 1.$$

For $p = 0.6$: $\mathbb{E}[X_i] = 0.2$. After n steps:

$$\mathbb{E}[S_n] = n(2p - 1).$$

This is a systematic *drift* — the random walk now has a trend.

Key Takeaway

Key concepts: stochastic process, expected value, variance, standard deviation, the $\sqrt{n}$ growth of randomness.

3 Act 2 — The Gambling Game and the Martingale

3.1 The Simulation

Simulation

The same random walk, relabelled. The vertical axis now shows "€ profit or loss." A bold horizontal line marks zero — break-even.
Probability slider: $p > 0.5$ gives the player an edge; $p < 0.5$ gives the house an edge.
Doubling strategy mode: Runs 10,000 gamblers using the martingale betting strategy. A histogram shows the distribution of final wealth — many small winners, a few catastrophic losers. A counter shows average profit converging to zero.
Build your own stopping rule: Users define a rule ("Stop when ahead by €K"). The simulation runs 10,000 gamblers. Average winnings at stopping converge to zero.

3.2 Intuitive Explanation

Intuitive Explanation

Win €1 for heads, lose €1 for tails. Cumulative winnings trace the same random walk from Act 1.
The gambler's temptation: "I'll keep playing until I'm ahead, then quit." Sounds foolproof — surely a run of luck will eventually put you in the green?
Mathematically, a fair random walk does eventually return to positive territory. But the time it takes can be astronomically long, and while you wait you might lose everything.
Central insight: There is no strategy that improves your expected outcome in a fair game. This is a proven mathematical theorem. Try any stopping rule in the simulation — the average winnings always converge to zero.

Intuitive Explanation

The doubling strategy (Martingale betting), exposed.
Lose, double the bet, repeat until you win back your losses plus €1. It works — most of the time.
But a long losing streak forces bets of €1, 2, 4, 8, 16, 32, 64, ... After just 10 consecutive losses, you must bet €1,024 to win back €1. The histogram makes it vivid: a tall bar of people who won €1, and a barely visible bar at the far left of people who lost thousands. Multiply heights by positions and sum: the average is zero.

3.3 Mathematical Formulation

Definition 1 (Martingale)**.** *A sequence $M_0, M_1, M_2, \ldots$ is a **martingale** if, at every step, the best prediction of the next value is the current value:*

$$\boxed{\mathbb{E}\Big[M_n \mid M_0, M_1, \ldots, M_{n-1}\Big] = M_{n-1}}$$

Reading the notation carefully:

- $\mathbb{E}[\cdot]$ = "the expected value of…" (long-run average).
- The vertical bar " | " = "given that we know." Everything after the bar is information we have.
- $M_0, M_1, \ldots, M_{n-1}$ = the full history up to step $n-1$.
- Full sentence: "The expected value of M_n, given that we know everything up to step $n-1$, equals M_{n-1}."

> Knowing the entire history, your best guess for the next value is just the current value. The past gives no useful information about the future direction.

Verifying: the coin-flip game is a martingale

Let $M_n = S_n$ (cumulative winnings). Then $M_n = M_{n-1} + X_n$.

$$\begin{aligned}
\mathbb{E}[M_n \mid M_0, \ldots, M_{n-1}] &= \mathbb{E}[M_{n-1} + X_n \mid M_0, \ldots, M_{n-1}] \\
&= M_{n-1} + \mathbb{E}[X_n \mid M_0, \ldots, M_{n-1}] && (M_{n-1} \text{ is known, passes through } \mathbb{E}) \\
&= M_{n-1} + 0 && (\text{next flip is independent; } \mathbb{E}[X_n] = 0) \\
&= M_{n-1}. \quad \checkmark
\end{aligned}$$

Doob's Optional Stopping Theorem (1953)

Theorem 1 (Doob's Optional Stopping Theorem)**.** *Let $\{M_n\}$ be a martingale and τ a **stopping time** — a rule for when to quit that uses only information accumulated so far (no looking into the future). Then, under reasonable conditions:*

$$\boxed{\mathbb{E}[M_\tau] = \mathbb{E}[M_0]}$$

> For a gambler starting with $M_0 = 0$: expected winnings at stopping = 0, always, for any strategy. **You cannot beat a fair game by choosing when to quit.**

What is a stopping time?

τ is a decision rule of the form: "Stop after step n if [condition based only on what happened so far]."

Rule	**Valid?**
Stop after exactly 100 flips	✓
Stop the first time I am ahead by €10	✓
Stop when I have lost €50	✓
Stop after 1000 flips *or* when ahead by €5, whichever first	✓
Stop right before the next loss	× (requires the future!)

The "reasonable conditions" for Doob's theorem

Doob's theorem requires at least one of:

1. $\tau \leq N$ for some fixed N (bounded stopping time), **or**
2. $|M_n| \leq C$ for some fixed C (bounded martingale), **or**
3. $\mathbb{E}[\tau] < \infty$ *and* certain integrability conditions (finite expected stopping time).

When *none* hold — as with the doubling strategy — the guarantee breaks down.

Why the doubling strategy fails

The doubling strategy guarantees eventual profit *only if* you have infinite money and infinite time. In practice:

- Finite budget $\Rightarrow$ real probability of ruin before recovery.
- Expected time to recovery is infinite: $\mathbb{E}[\tau] = \infty$.
- Doob's theorem gives $\mathbb{E}[M_\tau] \leq 0$, not $= 0$.

Why this matters for what is coming

The martingale concept reappears in every subsequent act. The likelihood ratio (Act 4) is a martingale. The mixture likelihood ratio (Act 7) is a martingale. The reason sequential tests control false positives — even with peeking — is that they exploit the same structure: **a fair game cannot be beaten by choosing when to stop.**

Key Takeaway

Key concepts: martingale, conditional expectation, stopping time, Doob's Optional Stopping Theorem, the "reasonable conditions."

4 Act 3 — Hypothesis Testing: Telling Two Coins Apart

4.1 The Simulation

Simulation

The user is presented with a coin and must decide: fair or biased?
Two panels run side by side:

- **Left:** Coin is actually fair ($p = 0.5$).
- **Right:** Coin is actually biased ($p = 0.5 + \delta$).

After each flip, a running tally shows heads/tails and current proportion. For small δ, the sequences look nearly identical for many flips.
User controls: true bias δ; number of flips before deciding.

4.2 Intuitive Explanation

Intuitive Explanation

You suspect a coin might be unfair. Flip it many times and look at the results. 10 flips with 6 heads? Could be luck. 1,000 flips with 600 heads? Strong evidence. Statisticians frame this as two competing stories:

- **Null hypothesis H_0:** "The coin is fair. Nothing is happening."
- **Alternative hypothesis H_1:** "The coin is biased. Something real is going on."

Two types of mistakes:

1. **False positive** (Type I): conclude biased when it is fair.
2. **False negative** (Type II): conclude fair when it is biased.

Traditionally: $\alpha = 0.05$ (max false positive rate), $\beta = 0.20$ (max false negative rate). **Power** $= 1 - \beta = 0.80$.

4.3 Mathematical Formulation

The likelihood

Suppose n flips produce k heads.

Under H_0 (fair coin, $p = 0.5$):

$$\mathcal{L}_0 = (0.5)^k \times (0.5)^{n-k} = (0.5)^n.$$

Under H_1 (biased coin, $p = 0.5 + \delta$):

$$\mathcal{L}_1 = (0.5 + \delta)^k \times (0.5 - \delta)^{n-k}.$$

The likelihood ratio

$$\boxed{\Lambda_n = \frac{\mathcal{L}_1}{\mathcal{L}_0} = \frac{(0.5 + \delta)^k \, (0.5 - \delta)^{n-k}}{(0.5)^n}}$$

"How many times more likely is our observed data under the biased-coin story compared to the fair-coin story?"

Worked example. $\delta = 0.1$ (biased coin has $p = 0.6$); $n = 10$ flips, $k = 7$ heads.

$$\begin{aligned}
\mathcal{L}_1 &= (0.6)^7 \times (0.4)^3 = 0.0280 \times 0.064 \approx 0.001792, \\
\mathcal{L}_0 &= (0.5)^{10} \approx 0.000977, \\
\Lambda_{10} &= \frac{0.001792}{0.000977} \approx 1.83.
\end{aligned}$$

The data is $\approx$1.83 times more likely if the coin is biased. Evidence, but not overwhelming.

Interpreting Λ_n

$\Lambda_n = 1$	Data equally consistent with both hypotheses
$\Lambda_n = 10$	$10\times$ more likely under H_1 — moderate evidence
$\Lambda_n = 100$	$100\times$ more likely under H_1 — strong evidence
$\Lambda_n = 0.1$	$10\times$ more likely under H_0 — evidence for fairness

Incremental updating

Λ_n can be computed one flip at a time:

$$\boxed{\Lambda_n = \Lambda_{n-1} \times \frac{f_1(x_n)}{f_0(x_n)}}$$

where $f_j(x_n)$ is the probability of the nth flip's outcome under H_j.

- Heads: multiply by $\dfrac{0.5 + \delta}{0.5}$.
- Tails: multiply by $\dfrac{0.5 - \delta}{0.5}$.

This incremental structure is what makes *sequential* testing possible.

Starting value

Before any data:

$$\Lambda_0 = 1.$$

Neither hypothesis is favoured. This will be important for Ville's inequality (which requires the starting value to equal 1).

Key Takeaway

Key concepts: null hypothesis, alternative hypothesis, likelihood, likelihood ratio, false positive, false negative, power, incremental updating.

5 Act 4 — The Likelihood Ratio Is a Martingale

5.1 The Simulation

Simulation

The likelihood ratio Λ_n is now plotted as a path over time — like the random walk from Act 1.
Left panel: 100 paths of Λ_n when the coin is truly *fair* (H_0 true). Paths wander with no systematic trend.
Right panel: 100 paths when the coin is truly *biased* (H_1 true). Paths drift upward.

5.2 Intuitive Explanation

Intuitive Explanation

Under H_0, the likelihood ratio wanders up and down but never develops a consistent trend. On average, it stays at 1.
This *is* a martingale — exactly like the gambler's cumulative winnings. Everything from Act 2 applies: Doob's theorem says you cannot systematically make Λ_n large by choosing clever peeking times.

5.3 Mathematical Formulation

Proof that Λ_n is a martingale under H_0

We must show: $\mathbb{E}[\Lambda_n \mid \Lambda_0, \ldots, \Lambda_{n-1}] = \Lambda_{n-1}$.

From Act 3, $\Lambda_n = \Lambda_{n-1} \times \dfrac{f_1(x_n)}{f_0(x_n)}$. Therefore:

$$\begin{aligned}
\mathbb{E}\Big[\Lambda_n \mid \text{past}\Big] &= \mathbb{E}\left[\Lambda_{n-1} \cdot \frac{f_1(x_n)}{f_0(x_n)} \,\middle|\, \text{past}\right] \\
&= \Lambda_{n-1} \cdot \mathbb{E}\left[\frac{f_1(x_n)}{f_0(x_n)}\right] && (\Lambda_{n-1} \text{ is known}) \qquad (1)
\end{aligned}$$

Now the key step. Under H_0, x_n is drawn from f_0. So:

$$\begin{aligned}
\mathbb{E}\left[\frac{f_1(x_n)}{f_0(x_n)}\right] &= \sum_{\text{all outcomes } x} f_0(x) \cdot \frac{f_1(x)}{f_0(x)} \\
&= \sum_x \cancel{f_0(x)} \cdot \frac{f_1(x)}{\cancel{f_0(x)}} && \textbf{(the cancellation!)} \\
&= \sum_x f_1(x) \\
&= 1. && (f_1 \text{ is a probability distribution; probabilities sum to 1}) \qquad (2)
\end{aligned}$$

Therefore:

$$\mathbb{E}\Big[\Lambda_n \,|\, \text{past}\Big] = \Lambda_{n-1} \times 1 = \Lambda_{n-1}. \qquad \checkmark$$

The cancellation with actual numbers

For $\delta = 0.1$ (biased coin $p = 0.6$):

Outcome	$f_0(x)$	$f_1(x)/f_0(x)$	$f_0(x) \times \big[f_1(x)/f_0(x)\big]$
Heads	0.5	$0.6/0.5 = 1.2$	$0.5 \times 1.2 = 0.6$
Tails	0.5	$0.4/0.5 = 0.8$	$0.5 \times 0.8 = 0.4$
		Total:	**1.0** ✓

When we compute the expected ratio under H_0, the H_0 probabilities cancel out, and we are just left summing H_1 probabilities — which must add to 1. This is the "aha" moment of the proof.

Two additional properties

1. Λ_n **is non-negative:** It is a product of non-negative ratios (probabilities divided by probabilities), so $\Lambda_n \geq 0$.
2. $\Lambda_0 = 1$**:** Before any data, neither hypothesis is favoured.

So Λ_n is a **non-negative martingale starting at 1**. Remember these properties — they are exactly what Ville's inequality needs.

Under H_1, Λ_n drifts upward

When the coin really is biased:

$$\mathbb{E}\left[\frac{f_1(x_n)}{f_0(x_n)}\right]_{H_1} = \sum_x f_1(x) \cdot \frac{f_1(x)}{f_0(x)} > 1.$$

So Λ_n tends to grow — this upward drift is the signal the test detects.

Key Takeaway

Key concepts: the likelihood ratio is a martingale under H_0; the cancellation that makes it work; non-negative martingale; drift under H_1.

6 Act 5 — Markov, Ville, and the Anytime-Valid Guarantee

6.1 The Simulation

Simulation

This is the pivotal simulation of the entire presentation.
Panel 1 — Markov (single time): 10,000 fair-coin paths of Λ_n. A vertical line at $n = 200$. A threshold at $\Lambda = 1/\alpha = 20$ (for $\alpha = 0.05$). Count: how many paths are above the threshold *at step 200*? Fraction $\leq 5\%$.
Panel 2 — Ville (any time): Same 10,000 paths. Count: how many *ever* crossed the threshold at *any* step? Also $\leq 5\%$ — but requires a stronger theorem.
Panel 3 — Comparison: 10,000 paths of a standard z-score checked at every step. Count how many ever show significance. Much higher than 5% — the peeking problem.
Summary:

- Markov: controls false positives at a single time ✓
- Ville: controls across all times simultaneously ✓
- Standard test with peeking: does NOT control ×

6.2 Intuitive Explanation

Intuitive Explanation

We are about to prove the most important result in this entire presentation: the **anytime-valid guarantee**.
Two steps:

1. **Markov's inequality** — a simple bound at one point in time.
2. **Ville's inequality** — extends the bound to all points in time simultaneously.

Ville's inequality is *the reason* sequential testing works.

6.3 Mathematical Formulation

Step 1: Markov's Inequality

Theorem 2 (Markov's Inequality)**.** *For any non-negative random variable X with finite $\mathbb{E}[X]$:*

$$\boxed{\mathbb{P}(X \geq c) \;\leq\; \frac{\mathbb{E}[X]}{c}}$$

$\mathbb{P}(\cdot)$ denotes probability — a number between 0 and 1 measuring how likely an event is. So "the probability that X is at least c cannot exceed the average of X divided by c."

Why is this true? An intuitive proof.

Suppose $X \geq 0$ always. Let $q = \mathbb{P}(X \geq c)$ be the fraction of outcomes at or above c. Those outcomes contribute *at least* $q \times c$ to the average (each is $\geq c$). The rest contribute ≥ 0 (since $X \geq 0$):

$$\mathbb{E}[X] \;\geq\; q \cdot c + (1-q) \cdot 0 = q \cdot c \qquad \Longrightarrow \qquad q \leq \frac{\mathbb{E}[X]}{c}. \qquad \checkmark$$

Concrete example. Average income in a town is €50,000.

$$\mathbb{P}(\text{income} \geq €500{,}000) \;\leq\; \frac{50{,}000}{500{,}000} = 0.10 = 10\%.$$

At most 10% of people can earn €500,000+.

Applying Markov to the likelihood ratio.

Under H_0, Λ_n is a martingale with $\mathbb{E}[\Lambda_n] = \Lambda_0 = 1$ and $\Lambda_n \geq 0$. Markov gives:

$$\mathbb{P}\left(\Lambda_n \geq \frac{1}{\alpha}\right) \leq \frac{\mathbb{E}[\Lambda_n]}{1/\alpha} = 1 \cdot \alpha = \alpha.$$

At any single, pre-specified step n, the false positive probability is at most α. Reassuring — but only for *one* moment.

Step 2: Ville's Inequality (1939)

Theorem 3 (Ville's Inequality)**.** *For any* ***non-negative martingale*** $\{M_n\}_{n\geq 0}$ *with* $M_0 = m_0$*:*

$$\boxed{\mathbb{P}\left(\sup_{n\geq 0} M_n \;\geq\; c\right) \;\leq\; \frac{m_0}{c}}$$

where $\sup_{n\geq 0} M_n$ *is the highest value* M_n ever *reaches.*

Why this is remarkable:

Markov	At any single *fixed* moment, $\mathbb{P}(M_n \geq c)$ is small.
Ville	$\mathbb{P}(M_n$ **ever** $\geq c$, at **any** moment) is *also* small.

Markov would allow the process to spike above c at various times, as long as it is rarely there at any single pre-specified time. Ville says even those spikes are collectively rare.

The intuition behind Ville. For a non-negative martingale, reaching a high value "uses up" the available expected value. Once high, it must stay high on average (martingale property), yet it cannot go below zero (non-negativity). These two constraints together limit how often it can ever be high.

Applying Ville to the likelihood ratio

Set $M_n = \Lambda_n$, $m_0 = \Lambda_0 = 1$, $c = 1/\alpha$:

$$\boxed{\mathbb{P}\left(\Lambda_n \text{ ever reaches } \frac{1}{\alpha} \text{ or higher}\right) \;\leq\; \alpha}$$

The probability that the likelihood ratio ever crosses our rejection threshold — at any point, across all flips, even if you run forever — is at most α.
This is the anytime-valid guarantee. You can peek after every observation. The false positive rate, across all checks combined, stays $\leq \alpha$.

Connecting back to Act 0

Standard test statistics (like p-values from a t-test) are *not* martingales. Checking repeatedly gives many independent chances to be fooled.

The likelihood ratio *is* a martingale under H_0. Ville's inequality converts this into a practical guarantee: **peeking is safe.**

Historical Note

Jean Ville proved this inequality in his 1939 doctoral thesis, *Étude critique de la notion de collectif*. It predated the formal development of martingale theory by Doob (1953) and was largely overlooked for decades. Robbins (1970) was among the first to recognise its practical importance for sequential testing — a connection we explore in Act 7.

Key Takeaway

Key concepts: Markov's inequality (with proof), Ville's inequality, the anytime-valid guarantee, martingale + non-negative = peek-safe.

7 Act 6 — SPRT: Wald's Sequential Probability Ratio Test (1945)

7.1 The Simulation

Simulation

Two coins: one fair (H_0), one biased (H_1). The user does not know which they have.
Λ_n is plotted as a path. Two thresholds:

- **Upper** at $\Lambda_n = B$: cross this $\Rightarrow$ "coin is biased."
- **Lower** at $\Lambda_n = A$: fall below this $\Rightarrow$ "coin is fair."

The path drifts toward one boundary and eventually crosses.
User controls: which coin is used, α, β, δ.
Additional panel: Distribution of stopping times across 10,000 simulated experiments, alongside the fixed-sample size required for the same α and β. The average SPRT stopping time is visibly less.

7.2 Intuitive Explanation

Intuitive Explanation

Abraham Wald (1943, published 1945) asked: given that we can peek at the data anytime, what is the most *efficient* way to decide between two hypotheses?
His answer: compute Λ_n after each observation. If it gets high enough, conclude H_1. Low enough, conclude H_0. Otherwise, keep collecting.
Wald proved this is **optimal**: no other sequential test with the same error guarantees needs fewer observations on average. In fact, the SPRT typically requires **47–63% fewer observations** than the best fixed-sample test with the same error rates.

7.3 Mathematical Formulation

Decision rule

Choose error tolerances α (false positive rate) and β (false negative rate). Compute two thresholds:

$$\boxed{B = \frac{1-\beta}{\alpha}}, \qquad \boxed{A = \frac{\beta}{1-\alpha}}.$$

Example ($\alpha = 0.05$, $\beta = 0.20$):

$$B = \frac{0.80}{0.05} = 16, \qquad A = \frac{0.20}{0.95} \approx 0.211.$$

Condition	Decision	Meaning
$\Lambda_n \geq B = 16$	Reject H_0	"Coin is biased"
$\Lambda_n \leq A \approx 0.21$	Accept H_0	"Coin is fair"
$A < \Lambda_n < B$	Keep flipping	"Not enough evidence yet"

Where the thresholds come from (Wald's inequalities)

These are **Wald's approximations**, derived from fundamental inequalities he proved in his 1945 paper. Let $\alpha' = \mathbb{P}_{H_0}(\text{reject } H_0)$ be the actual Type I error and $\beta' = \mathbb{P}_{H_1}(\text{accept } H_0)$ the actual Type II error. Wald showed:

$$\alpha' \leq \frac{1}{B},$$
$$\beta' \leq A.$$

Setting $B = (1-\beta)/\alpha$ and $A = \beta/(1-\alpha)$ guarantees $\alpha' \leq \alpha/(1-\beta)$ and $\beta' \leq \beta/(1-\alpha)$. When α and β are small (the typical case), these are very close to α and β themselves.

> The thresholds are set so that if the likelihood ratio ever climbs to B or drops to A, we can be confident in our decision. The approximation is slightly conservative — actual error rates are at or below the targets.

The random walk view

Equivalently, define the log-likelihood ratio $Z_n = \log \Lambda_n = \sum_{i=1}^{n} \log \frac{f_1(x_i)}{f_0(x_i)}$, where log denotes the natural logarithm (the inverse of the exponential function: if $e^y = x$ then $\log x = y$). The SPRT becomes a random walk between two barriers: stop at the first n such that $Z_n \geq \log B$ or $Z_n \leq \log A$.

Expected sample size (Wald's ASN formula)

Using what is now called **Wald's equation** — $\mathbb{E}[Z_n] = \mathbb{E}[n] \cdot \mathbb{E}[z]$, where $z = \log \frac{f_1(x)}{f_0(x)}$ is the single-observation log-likelihood ratio — Wald derived:

$$\boxed{E_0[n] \approx \frac{(1-\alpha)\log A + \alpha \log B}{\mathbb{E}_0[z]}}$$

under H_0, and

$$\boxed{E_1[n] \approx \frac{\beta \log A + (1-\beta)\log B}{\mathbb{E}_1[z]}}$$

under H_1.

> The expected number of observations depends on the log-thresholds ($\log A$ and $\log B$) and how much information each observation provides (measured by $\mathbb{E}[z]$, which depends on how different H_0 and H_1 are).

Savings over fixed-sample tests

Wald computed the fixed-sample size needed for the same α and β, and compared it to the SPRT's expected sample size. His results (Table 1 in the 1945 paper):

α	β	**Saving in expected observations**
0.05	0.05	$\approx 53\%$
0.05	0.01	$\approx 47\%$
0.01	0.01	$\approx 63\%$

The SPRT needs, on average, only about **half as many observations** as a traditional fixed-sample test. This was a remarkable result in 1945 and remains one of the main motivations for sequential testing.

Optimality of the SPRT

Historical Note

Wald conjectured in his 1945 paper (footnote 12, p. 157) that the SPRT is *exactly* optimal, not merely approximately so. He proved it only for the special case where the log-likelihood ratio jumps in fixed increments (zero overshoot). Three years later, Wald & Wolfowitz (1948) proved the general result: among **all** sequential tests (not just SPRTs) with the same or smaller error probabilities (α, β), the SPRT minimises the expected sample size under both H_0 and H_1 simultaneously.

Truncation

In practice, experiments cannot run forever. Wald showed that truncating the SPRT at a maximum number of observations n_0 barely affects error rates. For example, with $n_0 = 2N$ (where N is the fixed-sample size), the increase in α and β is negligible.

The critical weakness

SPRT requires specifying δ in advance — the exact alternative hypothesis H_1. If you set $\delta = 0.1$ but the true bias is 0.03, the likelihood ratio is computed against the wrong alternative. The test may take much longer, or the test may still determine the correct answer but far less efficiently.

In A/B testing — where the effect size is typically unknown — this is a serious limitation. The next two acts address this.

Key Takeaway

Key concepts: SPRT decision rule, Wald's boundary approximations, Wald's equation for expected sample size, 47–63% saving over fixed-sample tests, optimality (Wald & Wolfowitz 1948), the critical dependence on a pre-specified effect size.

8 Act 7 — The Mixture Approach: From Robbins (1970) to the mSPRT

This act has two parts. First, the key idea: mixing over effect sizes. Second, the application to A/B testing by Johari et al. (2017).

8.1 The Simulation

Simulation

Panel 1 — The problem with wrong δ: Three simultaneous paths: SPRT with the correct δ, SPRT with a δ that is too large, and SPRT with a δ that is too small. The correct one is fastest; the wrong ones are slow or stuck.
Panel 2 — The mixture idea: Like Act 6, but the user no longer specifies δ. A new slider controls τ (width of a "prior belief" over effect sizes). Λ_n^H is plotted. Only the *upper* threshold $1/\alpha$ is shown (no lower boundary).
Panel 3 — Comparison: A power curve and average stopping time chart comparing SPRT (with correct δ) to the mSPRT across a range of true effect sizes. The mSPRT is slightly less efficient at any single δ but much more robust overall.

8.2 Part A: Robbins' Mixture Approach (1970)

Intuitive Explanation

Intuitive Explanation

Wald's SPRT is like betting everything on one horse: you pick a specific effect size δ and compute evidence based on that choice. If the horse wins (the true effect is near δ), you are fast. Otherwise, slow.
Herbert Robbins proposed a radical alternative in 1970: **bet on all horses simultaneously.** Instead of choosing one δ, spread your "evidence budget" across many possible effect sizes using a weighting scheme (a mixing distribution H).
The key insight: this average likelihood ratio is *still* a non-negative martingale under H_0. Ville's inequality still applies. The anytime-valid guarantee is preserved.

The fundamental result

Robbins built on Ville's inequality directly. He observed that if Λ_n^δ is the likelihood ratio for a specific effect size δ, then the **mixture likelihood ratio**

$$\boxed{\Lambda_n^H = \int \Lambda_n^\delta \, dH(\delta)}$$

is also a non-negative martingale starting at 1 under H_0, because:

1. Each Λ_n^δ is a non-negative martingale under H_0 (Act 4).
2. A weighted average (integral) of martingales is a martingale (linearity of conditional expectation).

Therefore Ville's inequality gives:

$$\mathbb{P}\left(\Lambda_n^H \text{ ever} \geq \frac{1}{\alpha}\right) \leq \alpha.$$

Symbol by symbol:

- Λ_n^δ = the likelihood ratio assuming effect size δ.
- $dH(\delta)$ = how much weight to give effect size δ.
- $\int$ = continuous version of summation — averages over all possible δ.

Think: "average likelihood ratio across an infinitely large expert panel, weighted by credibility."

Robbins' confidence sequences

Robbins went further. He showed that inverting the mixture test gives a **confidence sequence** — a sequence of intervals I_n that covers the true parameter at *all* sample sizes simultaneously:

$$\mathbb{P}\Big(\mu \in I_n \text{ for every } n \geq 1\Big) \geq 1 - \alpha.$$

For the normal case with observations $x_i \sim \mathcal{N}(\mu, 1)$ and $S_n = \sum_{i=1}^n x_i$, he derived the two-sided confidence sequence:

$$\boxed{I_n = \left(\frac{S_n - c_n}{n}, \frac{S_n + c_n}{n}\right)}$$

where

$$c_n = \sqrt{(n+m)\left(a^2 + \log\left(\frac{n}{m} + 1\right)\right)}$$

and $\mathbb{P}(\mu \notin I_n \text{ for some } n) \leq e^{-a^2}$. Here $m > 0$ is a free parameter and a^2 controls the coverage (e.g. $a^2 = 6$ gives $\geq 95\%$ simultaneous coverage).

Unlike a standard confidence interval (valid at one pre-specified time), a confidence sequence is valid at every time simultaneously. You can peek as often as you want. Robbins proved this using exactly the mixture likelihood approach — Ville's inequality does the heavy lifting.

Connection to the law of the iterated logarithm

Historical Note

Robbins showed that the *tightest possible* confidence sequence boundary grows as $\sqrt{2n \log\log n}$ — exactly the rate of the **law of the iterated logarithm** (LIL), which describes the furthest excursions of a random walk. Wider boundaries (like the $\sqrt{n \log n}$ rate from the Normal mixing distribution) give tighter probability bounds but larger intervals at any finite n. The practical tradeoff: LIL-rate boundaries are asymptotically optimal but wider for moderate n; the Normal mixture boundaries are slightly wider asymptotically but tighter over the 2–3 orders of magnitude of sample sizes relevant to A/B testing.

8.3 Part B: The mSPRT for A/B Testing (Johari et al., 2017)

Intuitive Explanation

Intuitive Explanation

Johari, Pekelis, and Walsh (2017) applied Robbins' idea to the specific setting of A/B tests. They chose the Normal distribution as the mixing distribution H and derived a closed-form formula for the mixture likelihood ratio. They called it the **mixture Sequential Probability Ratio Test (mSPRT)**.
Their key contribution: making the theory practical. They gave explicit formulas, calibration guidance for τ, and proved that being within an order of magnitude of the true effect-size variance is sufficient for near-optimal power.

Closed form for Normal observations

Assumptions:

- Observations $x_i \sim \mathcal{N}(\mu, \sigma^2)$ (Normal with mean μ, **known** variance σ^2).
- Mixing distribution $H = \mathcal{N}(0, \tau^2\sigma^2)$ (Normal centred at 0, variance proportional to σ^2).

The mixture likelihood ratio has an exact closed form (Johari et al., Proposition 1):

$$\boxed{\Lambda_n^H = \frac{1}{\sqrt{1+n\tau^2}} \exp\left(\frac{n\tau^2\,\bar{x}_n^2}{2\,(\sigma^2/n)\,(1+n\tau^2)}\right)}$$

Unpacking every piece:

Symbol	Meaning
σ^2	Variance of individual observations (how noisy the data is).
n	Number of observations so far.
τ^2	Variance of the mixing distribution H, expressed as a multiple of σ^2. Controls how wide a range of effect sizes we consider plausible.
$\bar{x}_n$	Sample mean after n observations: $\bar{x}_n = \frac{1}{n}\sum_{i=1}^{n} x_i$.
$1/\sqrt{1+n\tau^2}$	Shrinkage factor. Slowly decreases as n grows; prevents the statistic from growing merely because more data arrived.
$\exp(\cdot)$	$e \approx 2.718$ raised to the argument. Grows rapidly when $\bar{x}_n$ is far from zero.

"The mSPRT score is large when the observed average is far from zero (strong signal), there are many observations, and the noise is low relative to the prior width."

Decision rule

$$\text{Stop and reject } H_0 \text{ when: } \boxed{\Lambda_n^H \geq \frac{1}{\alpha}}$$

For $\alpha = 0.05$: stop when $\Lambda_n^H \geq 20$.

No lower boundary. Unlike the SPRT, the mSPRT cannot definitively confirm H_0. The mixture over all effect sizes, including very small ones, means the statistic hovers near 1 indefinitely under H_0. In practice, experiments run for a fixed maximum duration; if the threshold is never crossed, conclude: "No significant effect detected."

The role of τ

τ is...	**Effect**	**Best for**
Small ($\sim$0.01)	Conservative, slow to reject	Very small effects
Large ($\sim$0.5)	Aggressive, fast for large effects	Large effects
Matched to true δ	Near-optimal	Good prior knowledge

Johari et al. (Theorem 3) proved that the optimal τ is chosen to minimise the expected stopping time for a given true effect size. In practice, getting *within an order of magnitude* of the true effect-size variance suffices for near-optimal power.

Key assumption: known variance

The mSPRT formula assumes σ^2 is known. For continuous data this is typically estimated; for binary data (e.g. click/no-click), Johari et al. note that the exact mSPRT results hold only approximately via the Central Limit Theorem. For large samples ($n \geq 100$, nearly always true in A/B testing), the approximation is excellent.

Always-valid p-values and confidence intervals

Johari et al. also defined **always-valid p-values**:

$$p_n = \min\left(1, \frac{1}{\Lambda_n^H}\right)$$

and **always-valid confidence intervals** by inverting Λ_n^H:

$$C_n = \left\{\mu : \Lambda_n^H(\mu) < 1/\alpha\right\}.$$

These enjoy the usual sequential guarantee: at any stopping time τ, the Type I error and coverage are controlled.

Key Takeaway

Key concepts: Robbins' mixture idea, the mSPRT as a weighted average of likelihood ratios, the closed-form Normal formula, the role of τ, no lower boundary, the known-variance assumption.
What's next: The mSPRT was a landmark — but it is not what Eppo uses. The next act introduces **confidence sequences** from Howard et al. (2021), a more general framework that Eppo adopted instead.

9 Act 8 — Confidence Sequences: The Modern Framework (Howard et al., 2021)

9.1 The Simulation

Simulation

Panel 1 — Confidence Sequence vs. Fixed CI: Data arrives one observation at a time. Two bands track the estimated mean:

- A **fixed 95% CI** (recomputed at each step) — shrinks as $1/\sqrt{n}$, but invalidated by peeking.
- A **confidence sequence** — wider at first, but valid at *all* times simultaneously.

A counter shows how often the true mean escapes each band (across 10,000 runs). For the fixed CI with peeking: $\gg 5\%$. For the CS: $\leq 5\%$.

Panel 2 — The Normal mixture boundary: The boundary function $u(v)$ plotted against the variance $v = n\sigma^2$. A slider for ν shows how changing ν shifts the "sweet spot" — the sample size where the boundary is tightest relative to the fixed-sample CI.

9.2 Intuitive Explanation

Intuitive Explanation

A confidence sequence is the natural answer to the peeking problem for estimation (not just testing).

A standard 95% confidence interval says: "We are 95% confident the true value is in this range *right now*." But if you check again tomorrow with new data, the guarantee resets. Check 100 times and you are almost guaranteed to be wrong at least once.

A confidence sequence says: "We are 95% confident the true value has been in this range *at every single time we have ever checked, and at every time we will ever check in the future*."

This is a strictly stronger guarantee. The price: the band is wider than a fixed CI at any single time. But in exchange, you are free to monitor continuously, stop whenever you want, and report at any time.

9.3 Mathematical Formulation

Definition

Definition 2 (Confidence Sequence)**.** *A sequence of intervals $(C_t)_{t\geq 1}$ is a $(1-\alpha)$-**confidence sequence** (CS) for a parameter μ if:*

$$\boxed{\mathbb{P}\Big(\mu \in C_t \text{ for all } t \geq 1\Big) \;\geq\; 1-\alpha}$$

The **entire sequence** of intervals simultaneously covers the true parameter with probability at least $1 - \alpha$. Not just at one time — at every time.

Compare with a standard CI, which only guarantees $\mathbb{P}(\mu \in C_n) \geq 1 - \alpha$ for a single fixed n.

The sub-ψ framework

Howard et al. build confidence sequences from **sub-ψ processes**, which generalise the concept of "observations bounded by a moment generating function." The key idea:

1. Define a process S_t (e.g. the running sum of centred observations).
2. Verify that the process satisfies a **sub-ψ condition**: the moment generating function of $S_t - S_{t-1}$, conditional on the past, is bounded by a known function ψ.
3. Construct a **supermartingale** from this bound.
4. Apply Ville's inequality to get the anytime-valid guarantee.

For sub-Gaussian observations (which includes all bounded random variables and all Normal random variables), the relevant condition is:

$$\mathbb{E}\Big[e^{\lambda(X_t - \mu)} \,|\, \text{past}\Big] \;\leq\; e^{\lambda^2\sigma^2/2} \qquad \text{for all } \lambda \in \mathbb{R}.$$

This says: the tails of each observation are no heavier than those of a Normal distribution with variance σ^2.

The sub-ψ condition is a way of saying "the data is not too wild." For Normal data or bounded data (like click rates), this condition is automatically satisfied. It allows the theory to work *without* assuming the data is exactly Normal — making it **nonparametric**.

The Normal Mixture Boundary (the key formula)

The centrepiece of Howard et al. is the **Normal mixture confidence sequence** (Proposition 5, equation 14 in their paper). For sub-Gaussian observations with intrinsic time (cumulative variance) $v_t = \sum_{i=1}^{t} \sigma_i^2$:

$$\boxed{u(v) = \sqrt{(v + \nu) \cdot \log \frac{v + \nu}{\nu \alpha^2}}}$$

The $(1 - \alpha)$-confidence sequence is:

$$C_t = \left(\bar{x}_t - \frac{u(v_t)}{t},\; \bar{x}_t + \frac{u(v_t)}{t}\right)$$

where $v_t = t\sigma^2$ for i.i.d. observations with variance σ^2.

Unpacking the boundary:

Symbol	Meaning
v	**Intrinsic time**: the cumulative variance of observations so far. For i.i.d. data: $v_t = t\sigma^2$. For heterogeneous data: $v_t = \sum_{i=1}^{t} \sigma_i^2$.
$\nu > 0$	**Tuning parameter**: controls *which sample sizes* the boundary is tightest for. Specifically, the boundary is approximately optimised for $v \approx \nu$. Larger ν = tighter for larger samples, wider for smaller samples.
α	Significance level (e.g. 0.05).
$\log(\cdot)$	Natural logarithm.
$\sqrt{(v+\nu) \cdot \log \frac{v+\nu}{\nu\alpha^2}}$	The boundary grows roughly as $\sqrt{v \log v}$ for large v. This is slightly wider than the $\sqrt{v}$ growth of a fixed-sample CI, but the extra $\sqrt{\log v}$ is the "price of peeking."

The boundary $u(v)$ tells you how far from zero the running sum S_t can drift before you can reject H_0. The parameter ν lets you tune where the boundary is tightest. If you expect to collect N observations total with variance σ^2, setting $\nu \approx N\sigma^2$ makes the boundary tightest around $n = N$.

How ν works

ν is...	Effect	Best for
Small	Tight for small samples, wide for large	Early stopping
Large	Wide for small samples, tight for large	Long experiments
$\approx N\sigma^2$	Near-optimal at sample size N	Known experiment length

Connection to Robbins and the mSPRT

Howard et al. explicitly credit Robbins (1970) as their starting point. Their Normal mixture boundary comes from choosing a *Normal mixing distribution* over the mean parameter — exactly Robbins' approach. The connection:

- Robbins (1970): introduced the mixture approach and confidence sequences, but only for known-variance Normal data.
- Johari et al. (2017): made it practical for A/B testing via the mSPRT, but still assumed known variance and parametric (Normal) data.
- **Howard et al. (2021): extended to nonparametric settings** via the sub-ψ framework. The boundary works for *any* sub-Gaussian data (not just Normal), the variance can be unknown (estimated from data), and the coverage guarantee is exact (non-asymptotic).

Growth rate comparison

Method	Half-width rate	Source
Fixed-sample CI	$\frac{\sigma}{\sqrt{n}} \cdot z_{\alpha/2}$	Central Limit Theorem
Normal mixture CS	$\frac{1}{n}\sqrt{(n\sigma^2+\nu)\log\frac{n\sigma^2+\nu}{\nu\alpha^2}}$	Howard et al. (2021)
LIL-rate CS	$\frac{\sigma}{\sqrt{n}} \cdot \sqrt{2\log\log(n\sigma^2)}$	Law of iterated logarithm

The Normal mixture CS is wider than the fixed CI by a factor of $\approx \sqrt{\log n}$ (the price of peeking). It is slightly wider than the LIL-rate CS asymptotically, but **tighter in practice** over the 2–3 orders of magnitude of sample sizes relevant to A/B testing.

Why Eppo chose this framework

The Normal mixture boundary is the formula at the heart of Eppo's sequential testing system (Schmit & Miller, 2024). To see why, consider what an A/B testing platform needs:

1. **Anytime-valid:** Experimenters will peek — the guarantee must hold at every moment. ✓ (Ville's inequality via the sub-ψ supermartingale.)
2. **No known variance required:** Real experiments have unknown, potentially changing variance. ✓ (The sub-ψ framework permits estimated variance; Eppo plugs in $\hat{\sigma}^2(t)$ at each time step.)
3. **No distributional assumption:** Metrics like revenue or session length are far from Normal. ✓ (Sub-Gaussian condition is satisfied by all bounded data and approximately by most practical metrics at scale.)
4. **Tunable tightness:** Different experiments have different expected durations. ✓ (The parameter ν lets Eppo set $\nu = M\hat{\sigma}^2$, where M is the expected total sample size, making the boundary tightest around the planned analysis time.)
5. **Produces a confidence interval:** Experimenters need to report "the lift is +3%± 1.5%," not just "significant / not significant." ✓ (The CS directly gives an interval $\bar{x}_t \pm u(v_t)/t$ at every time.)

In Acts 9–12, we will see how Eppo wraps this boundary into a complete pipeline: the confidence sequence is applied to *regression-adjusted* observations (generalised CUPED) and reported as a confidence interval around the **relative lift** $\mu_1/\mu_0 - 1$. The key formula in Act 10 —

$$\mathrm{CI}(t) = \hat{\tau}(t) \;\pm\; \hat{\sigma}_{\hat{\tau}}(t) \cdot \sqrt{\frac{n+\nu}{n} \cdot \log\frac{n+\nu}{\nu\alpha^2}}$$

— is just the Normal mixture boundary $u(v)$ from this act, with v replaced by the estimated cumulative variance $n\hat{\sigma}^2_{\hat{\tau}}(t)$ and the boundary divided by n to convert from the sum scale to the mean scale.

Key Takeaway

Key concepts: confidence sequences (uniform coverage over all times), the Normal mixture boundary $u(v) = \sqrt{(v+\nu)\log\frac{v+\nu}{\nu\alpha^2}}$, the tuning parameter ν, the sub-ψ framework, nonparametric validity, the $\sqrt{\log n}$ "price of peeking."
This is the framework Eppo uses. The boundary formula from this act is the engine inside Eppo's confidence interval.

10 Act 9 — The Problem Eppo Solves

10.1 The Simulation

Simulation

What is being simulated: 10,000 parallel A/B experiments, each with a small true treatment effect, comparing classical and sequential approaches.

Panel 1 — Classical A/B test with peeking (failure case): Two groups (control / treatment) drawn from distributions with a small true difference. A classical 95% CI is recomputed at each time step.

10,000 parallel experiments. Counters show:

- "False conclusions if you stop when CI excludes 0": far above 5%.
- "Experiments still running after planned duration": many finish early with a wrong answer.

False decision visualisation: Experiments that stop early with the *wrong* conclusion are highlighted in red. A running counter displays **"Wrong winners: k / N"** — making it viscerally clear how many experiments would ship the wrong feature.

Panel 2 — Sequential CI (preview): Same experiments, but now using a sequential confidence interval (from Act 8). The band is wider at first, but the false conclusion rate stays $\leq 5\%$ regardless of when you peek. Incorrectly-stopped experiments are again highlighted, but the count stays near the α level.

Panel 3 — Time-to-detection comparison: Two versions of the same experiment side-by-side:

- **Without variance reduction (red):** Raw outcomes. CI is wide. Takes many observations to detect a real effect.
- **With variance reduction (blue):** Adjusted outcomes (noise removed). CI is narrow. Same effect detected much sooner.

A counter displays: **"Average samples to detect effect: n_{raw} vs. n_{adjusted}"** — quantifying the speed-up.

Controls:

- Effect size slider δ (0 to 10%).
- Variance level slider (low / medium / high).
- Number of parallel experiments (100 / 1,000 / 10,000).
- Fixed seed toggle (deterministic replay).

Key takeaway: Classical CIs break under peeking; sequential CIs stay valid. Variance reduction dramatically speeds up detection.

10.2 Intuitive Explanation

Intuitive Explanation

A real A/B testing platform faces three problems simultaneously:

1. **The peeking problem** (Act 0): Product managers check results daily. Classical tests break under repeated checking.
2. **High variance:** Users are wildly different. Some buy €0, others buy €500. This noise drowns out the signal — a 2% lift buried under 200% noise. Experiments take weeks to reach significance.
3. **Heterogeneous users:** Not all users respond the same way. Heavy spenders, occasional browsers, mobile vs. desktop — all mixed together. Covariates (pre-experiment behaviour, device type, country) carry information that can separate signal from noise.

What Eppo wants:

- **Anytime-valid inference:** Peek as often as you like; the error guarantee holds.
- **Faster experiments:** Use pre-experiment data to reduce noise, shrinking the confidence interval.
- **Simple outputs:** Report a confidence interval for the *relative lift* ("+3% ± 1.5%") that experimenters can act on immediately.

Acts 0–8 built each ingredient. The next three acts show how Eppo assembles them into one pipeline.

Key Takeaway

Three problems, three solutions:

Problem	Solution	Where we learned it
Peeking inflates errors	Sequential CI (Howard et al.)	Acts 5, 8
High variance / slow experiments	Regression adjustment (CUPED)	Act 11
Need interpretable output	Relative lift with CI	Act 12

11 Act 10 — The Eppo Pipeline, Step by Step (Schmit & Miller, 2024)

This is the central act of the Eppo section. It walks through the entire pipeline from data collection to decision, one step at a time.

11.1 The Simulation

Simulation

What is being simulated: A single A/B experiment flowing through Eppo's pipeline, stage by stage.
Animated pipeline diagram. Data flows left-to-right through labelled stages. Each stage lights up as it is explained.
Stages (rendered as connected boxes):

$$\boxed{\text{Randomise}} \rightarrow \boxed{\text{Collect}} \rightarrow \boxed{\text{Adjust}} \rightarrow \boxed{\text{Estimate}} \rightarrow \boxed{\text{CI}} \rightarrow \boxed{\text{Decide}}$$

Step-by-step mode: A button labelled **"Step through pipeline"** pauses at each stage. At each stage, a distinct visual state is shown:

Stage	Visual state
Randomise	Users appear as dots, randomly coloured blue (control) or green (treatment).
Collect	Each dot acquires labels: a pre-experiment value X and an outcome Y. Tooltip on hover shows both values.
Adjust	Dots animate: $Y \rightarrow Y^*$. A regression line overlays each group. Dots shift vertically to their residual position. Spread visibly decreases.
Estimate	Group means appear as horizontal lines. The lift calculation ($\hat{\tau}_{\text{rel}}$) displays as a labelled gap between lines.
CI	A confidence interval band draws around the estimated lift. The band narrows as more data arrives.
Decide	The CI is colour-coded: **green** (entirely above 0), **red** (entirely below 0), or **grey** (crosses 0). A verdict label appears.

Toggle: "With regression adjustment" / "Without regression adjustment." When toggled off, the "Adjust" box is greyed out, dots do not shift, and the CI is visibly wider.
Side-by-side comparison mode: An optional button labelled **"Compare with / without adjustment"** splits the display into two panels:

- **Left panel:** no regression adjustment (raw outcomes).
- **Right panel:** with regression adjustment (residuals).

Both panels use the *same underlying data* — only the adjustment differs. The CI width difference is immediately visible, making the benefit of variance reduction obvious at a glance.
Stage highlights (step-by-step mode): When stepping through the pipeline, each stage displays a **"Before / After"** mini-panel showing the data state entering and leaving that stage. Key changes are highlighted:

- **Adjust stage:** scatter shifts from raw to residual; variance annotation updates.
- **Estimate stage:** group means and lift label appear.
- **CI stage:** band draws with colour coding (green / red / grey).

Controls:

- True effect size δ (0 to 10%).
- Sample size n (100 to 100,000).

11.2 Intuitive Explanation

Intuitive Explanation

Think of Eppo's pipeline as a factory with six stations. Each station does one job. Data enters raw and exits as a trustworthy confidence interval.

11.3 The Pipeline

Step 1: Randomise

Each user arriving at the website is randomly assigned to **control** (existing experience) or **treatment** (new feature). Randomisation ensures the groups are comparable — any difference in outcomes is caused by the treatment, not by pre-existing differences.

Step 2: Collect data over time

As users interact with the product, two quantities are recorded for each user:

- Y_i: the **outcome** during the experiment (e.g. purchase amount, clicks, session length).
- X_i: a **pre-experiment covariate** — the same metric from before the experiment started (e.g. purchases in the prior 2 weeks), plus optionally other covariates (country, device type, etc.).

Data arrives continuously. The pipeline processes it in batches or continuously as it flows in.

Step 3: Regression adjustment (noise removal)

This is where Eppo *generalises* the CUPED idea from Deng et al. (2013).

For each group **separately**:

1. Fit a regression model predicting Y from the pre-experiment covariates X:
$$\hat{f}_{\text{control}}(X), \qquad \hat{f}_{\text{treatment}}(X)$$
2. Compute the **adjusted outcome** (residual) for each user:
$$Y_i^* = Y_i - \hat{f}_g(X_i)$$
where g is the user's group.

> "Predict what each user would have done based on their history, then subtract that prediction. What's left is the part you *couldn't* have predicted — and that's where the treatment effect hides."

Why separate models? The relationship between covariates and outcome might differ between groups (e.g. heavy users may respond more to the treatment). Fitting separately captures this.

Why this generalises CUPED: Classic CUPED uses a single linear coefficient θ applied identically to both groups. Eppo's approach allows nonlinear models and group-specific coefficients — strictly more general.

Step 4: Estimate the treatment effect

Compute the adjusted group means:

$$\hat{\mu}_0(t) = \frac{1}{n_0(t)} \sum_{i \in \text{control}} Y_i^*, \qquad \hat{\mu}_1(t) = \frac{1}{n_1(t)} \sum_{i \in \text{treatment}} Y_i^*$$

The estimated treatment effect:

$$\hat{\tau}(t) = \hat{\mu}_1(t) - \hat{\mu}_0(t)$$

And the **relative lift** (dividing by the control mean):

$$\boxed{\text{Relative lift}(t) = \frac{\hat{\tau}(t)}{\hat{\mu}_0(t)} = \frac{\hat{\mu}_1(t)}{\hat{\mu}_0(t)} - 1}$$

If control averages €100 and treatment averages €103, the relative lift is $103/100 - 1 = 0.03 = +3\%$.

Why relative lift?

Eppo reports the **relative** lift $\hat{\tau}(t)/\hat{\mu}_0(t)$ rather than the absolute difference $\hat{\tau}(t)$. Why?

Scenario	Control mean	Treatment mean	Lift
High-value metric	€100	€103	+3%
Low-value metric	€10	€13	+30%

Both scenarios have the same absolute difference (€3), but the business impact is very different. Relative lift captures this:

- **Scale invariance:** A 3% lift means the same thing whether the baseline is €10 or €10,000.
- **Comparability:** You can compare lift across experiments on different metrics (clicks, revenue, sessions) because the denominator normalises the scale.
- **Business alignment:** Product teams think in percentages — "this feature increased conversion by 2%" — not in raw units.

UI toggle: The dashboard supports a toggle between **"Show absolute difference"** and **"Show relative lift."** Both are derived from $\hat{\tau}(t)$; relative lift simply divides by the control mean.

Step 5: Estimate the variance

Compute the standard error of $\hat{\tau}(t)$ from the **residuals** Y_i^*:

$$\hat{\sigma}^2_{\hat{\tau}}(t) = \frac{s_0^2(t)}{n_0(t)} + \frac{s_1^2(t)}{n_1(t)}$$

where $s_g^2(t)$ is the sample variance of the adjusted outcomes in group g.

The variance is estimated from the data — not assumed known. This is a key difference from the mSPRT (Act 7), which required known variance. Howard et al.'s framework (Act 8) permits this.

Step 6: Construct the sequential confidence interval

Plug the estimated treatment effect and its standard error into the Normal mixture boundary from Act 8:

$$\boxed{\mathrm{CI}(t) = \hat{\tau}(t) \;\pm\; \hat{\sigma}_{\hat{\tau}}(t) \cdot \sqrt{\frac{n+\nu}{n} \cdot \log \frac{n+\nu}{\nu\alpha^2}}}$$

where $n = n_0(t) + n_1(t)$ is the total sample size and $\nu = M \cdot \hat{\sigma}^2$ is the tuning parameter (set from the expected total sample size M).

Step 7: Decide

What you see	Colour	Action
CI entirely above 0	Green	Positive effect. Ship the feature.
CI entirely below 0	Red	Negative effect. Revert.
CI crosses 0, experiment ongoing	Grey	Inconclusive. Keep collecting.
CI crosses 0, experiment ended	Grey	No significant effect. Decide pragmatically.

Because the CI is a confidence sequence (Act 8), **this decision is valid at whatever time it is made.** No need to pre-specify an analysis time.

Key Takeaway

The Eppo pipeline in one sentence: Randomise → collect Y and covariates X → regression-adjust (remove predictable noise) → estimate lift and variance → wrap in a sequential CI → decide when CI excludes zero.
Every step matters. The regression removes noise (Act 11 dives deeper). The sequential CI guarantees validity (Act 8). The relative lift provides the business answer.

12 Act 11 — Variance Reduction via Regression Adjustment

12.1 The Simulation

Simulation

What is being simulated: The effect of regression adjustment on variance and confidence interval width, comparing raw outcomes with adjusted outcomes.

Panel 1 — Why raw averages are noisy: Scatter plot of 1,000 users: x-axis = pre-experiment purchases, y-axis = experiment purchases. Strong positive correlation visible. Two horizontal lines show the raw group means (control in blue, treatment in green). The gap between them *is* the treatment effect — but the scatter around each line is enormous.

Panel 2 — Regression adjustment: Same scatter plot, but now a regression line is fitted per group. Each dot is replaced by its **residual** (vertical distance from the regression line). The residuals have much less spread. The adjusted group means are shown — same gap, much less scatter.

Panel 3 — Effect on the confidence interval: Two confidence bands evolving over time:

- **Red (raw):** Wide band. Slowly narrows.
- **Blue (adjusted):** Narrow band. Reaches significance much sooner.

A counter shows: "Observations saved: $X\%$" — how many fewer observations the adjusted version needs to reach the same width.

Controls:

- Correlation slider ρ (0 to 0.95): controls how predictable users are from their pre-experiment behaviour.
- True effect size δ (0 to 10%).
- Toggle: with/without adjustment.
- Fixed seed toggle (deterministic replay).
- "Run 1,000 experiments" mode (batch statistics).

Key takeaway: The higher the correlation ρ between pre-experiment and experiment data, the more variance is removed. At $\rho = 0.7$, the experiment runs $\approx 2\times$ faster.

12.2 Intuitive Explanation

Intuitive Explanation

Imagine weighing yourself before and after a meal. If every person in the study has a different baseline weight (50 kg to 120 kg), the raw "post-meal weight" is extremely variable. But the *change* (before → after) is much more consistent — a meal adds roughly the same weight regardless of the person.
Regression adjustment does the same thing. Each user's outcome Y is partly explained by their pre-experiment behaviour X (their "baseline weight"). The regression model predicts that baseline. Subtracting the prediction leaves the *change* — the part due to the treatment, not the person.
The result: less noise, same signal, tighter confidence intervals, faster decisions.

12.3 Mathematical Formulation

From CUPED to generalised regression adjustment

Recall CUPED (Deng et al., 2013) from the notation in Act 10, Step 3:

$$Y^* = Y - \theta(X - \mathbb{E}[X])$$

with optimal $\theta = \mathrm{Cov}(Y, X)/\mathrm{Var}(X)$ and variance reduction $\mathrm{Var}(Y^*) = \mathrm{Var}(Y)(1 - \rho^2)$, where $\rho = \mathrm{Cor}(Y, X)$ is the **correlation coefficient** between the outcome and the covariate (a number between -1 and $+1$ measuring how strongly they move together).

Eppo generalises this in two ways:

1. **Nonlinear models:** Instead of a single linear coefficient θ, fit a full regression model $\hat{f}(X)$ that can capture nonlinear relationships (e.g. random forests, gradient boosting).
2. **Separate models per group:** Fit $\hat{f}_{\text{control}}(X)$ and $\hat{f}_{\text{treatment}}(X)$ independently. This allows the treatment to change how covariates relate to outcomes.

The adjusted outcome for user i in group g:

$$\boxed{Y_i^* = Y_i - \hat{f}_g(X_i)}$$

> "Take what user i actually did. Subtract what the model predicted they would do based on their history. What's left is the residual — the *unpredictable* part."

Why this reduces variance

The variance of the residual Y^* is:

$$\mathrm{Var}(Y^*) = \mathrm{Var}(Y - \hat{f}(X)) = \mathrm{Var}(Y)(1 - R^2)$$

where R^2 is the coefficient of determination of the regression model — the fraction of variance in Y that is explained by X.

R^2	$1 - R^2$	**Variance remaining**	**Equivalent traffic multiplier**
0.10	0.90	90%	1.1×
0.25	0.75	75%	1.3×
0.50	0.50	50%	2×
0.60	0.40	40%	2.5×
0.75	0.25	25%	4×

With $R^2 = 0.50$, the regression explains half the variance. The remaining noise is halved — equivalent to running the experiment with twice as many users.

What covariate to use

Following Deng et al. (2013):

1. **The same metric from the pre-experiment period** is consistently the most powerful covariate.
2. A pre-period of ~ 2 **weeks** works well for experiments lasting 1–3 weeks (longer pre-periods have diminishing returns).
3. Additional covariates (device type, country, day-of-week of first visit) provide marginal gains.

Practical results (Deng et al., 2013, at Microsoft Bing):

- Queries-per-user, clicks-per-user: **45–52%** variance reduction ($\Rightarrow$ roughly 2× traffic multiplier).
- Revenue-per-user: $< 5\%$ reduction (low temporal correlation).

The critical safety rule

Key Takeaway

The covariate X must satisfy $\mathbb{E}[X^{(\text{treatment})}] = \mathbb{E}[X^{(\text{control})}]$. This is automatically guaranteed when X comes from the *pre-experiment* period (randomisation has not yet occurred).
NEVER use post-treatment covariates. Deng et al. demonstrated that using an in-experiment covariate (highly correlated with the outcome) produced **directionally opposite conclusions**, because the treatment itself changed X.

Why adjustment does not break the sequential guarantee

The confidence sequence (Act 8) is applied to the adjusted outcomes Y^*, not the raw Y. The guarantee holds because:

1. Covariates come from the pre-experiment period (independent of treatment).
2. Regression adjustment changes only the variance, not the mean treatment effect.
3. The sub-Gaussian condition (Act 8) is satisfied by the adjusted residuals.
4. Ville's inequality applies to the resulting supermartingale.

The adjustment is like using a better measuring instrument — more precise, but measuring the same thing. The sequential guarantee depends on the structure of the martingale, which is preserved.

Key Takeaway

Key concepts: regression adjustment as generalised CUPED, separate models per group, variance reduction factor $1 - R^2$, the safety rule (pre-experiment covariates only), adjustment preserves the sequential guarantee.

13 Act 12 — Sequential Confidence Intervals in Eppo

13.1 The Simulation

Simulation

What is being simulated: The behaviour of Eppo's sequential confidence interval versus a classical CI, with controls for the tuning parameter ν and regression adjustment.

Panel 1 — Classical vs. sequential CI over time: A single experiment with a true effect of +3%. Two bands evolve over time:

- **Dashed grey:** Classical 95% CI (recomputed at each step). Narrow but invalid under peeking.
- **Solid blue:** Eppo's sequential CI. Wider at first, then approaches the classical CI as n grows.

A counter shows the "sequential multiplier" — the factor that replaces $z_{\alpha/2} \approx 1.96$ in the classical CI. It starts high (e.g. 3.5) and slowly decreases.

Panel 2 — ν tuning: A slider for ν (or equivalently M, the expected total sample size).

- Small ν (small M): CI is tight early, wide late.
- Large ν (large M): CI is wide early, tight late.
- $\nu \approx M\hat{\sigma}^2$: sweet spot at the planned analysis time.

An animation shows the boundary curve $u(v)$ reshaping as ν changes.

Panel 3 — 10,000 experiments: Run 10,000 parallel experiments. Two counters:

- **Classical CI with peeking:** "Fraction that ever falsely excluded 0": $\gg 5\%$.
- **Sequential CI with peeking:** Same counter: $\leq 5\%$.

Controls:

- True effect size δ (0 for null, or a positive value).
- Expected sample size M (slider that sets ν).
- Significance level α.
- Toggle: with/without regression adjustment.
- Fixed seed toggle (deterministic replay).
- "Run 1,000 experiments" mode (batch statistics).

Key takeaway: The sequential CI is wider than the classical CI (the "price of peeking") but guarantees validity under continuous monitoring. The tuning parameter ν controls where the CI is tightest.

13.2 Intuitive Explanation

Intuitive Explanation

This act connects the pipeline from Act 10 to the final statistical output that an experimenter sees on their dashboard.
A standard 95% CI says: "Right now, the lift is $+3\% \pm 1.5\%$." But that guarantee is only valid at this exact moment. If you checked yesterday, and you check again tomorrow, the guarantee has already been violated.
Eppo's sequential CI says: "The lift is $+3\% \pm 2.1\%$, and this has been valid *at every moment since the experiment started.*"
The price: the interval is wider — the "±2.1%" instead of "±1.5%". This is the **cost of peeking**. But it is a small price for the freedom to check whenever you want and stop as soon as you are convinced.

Why the interval shrinks slower

A standard CI shrinks as $1/\sqrt{n}$: double your sample, the interval narrows by $\sqrt{2} \approx 1.41$. Eppo's sequential CI shrinks slower — roughly as $\sqrt{\log n/n}$ — because it must remain valid across *all* possible stopping times, not just the one you chose.

Think of a standard CI as a promise valid at one specific moment: "At $n = 10{,}000$, I'm 95% confident." A sequential CI makes a much stronger promise: "At every n from 1 to ∞, I'm 95% confident."
The extra $\sqrt{\log n}$ factor is the price of that stronger promise. As data accumulates, the penalty grows so slowly that it barely matters — $\sqrt{\log(10{,}000)} \approx 3.0$, while $\sqrt{\log(100{,}000)} \approx 3.4$ — but it never disappears entirely.

13.3 Mathematical Formulation

The formula, revisited

From the pipeline (Act 10, Step 6):

$$\boxed{\mathrm{CI}(t) = \hat{\tau}(t) \;\pm\; \hat{\sigma}_{\hat{\tau}}(t) \cdot \underbrace{\sqrt{\frac{n+\nu}{n} \cdot \log\frac{n+\nu}{\nu\alpha^2}}}_{\text{sequential multiplier}}}$$

Compare with a standard CI:

$$\mathrm{CI}_{\text{classical}}(t) = \hat{\tau}(t) \;\pm\; \hat{\sigma}_{\hat{\tau}}(t) \cdot \underbrace{z_{\alpha/2}}_{\approx 1.96}$$

The only difference is the **sequential multiplier** replacing the constant 1.96.

How the sequential multiplier behaves

Define:

$$m(n) = \sqrt{\frac{n+\nu}{n} \cdot \log \frac{n+\nu}{\nu\alpha^2}}$$

n	$m(n)$ (example)	$z_{\alpha/2}$	**Interpretation**
100	≈ 3.8	1.96	Early: sequential CI is $\sim$2$\times$ wider
1,000	≈ 2.8	1.96	Mid-experiment: gap narrowing
10,000	≈ 2.3	1.96	Late: only $\sim$17% wider
100,000	≈ 2.1	1.96	Approaching classical, never quite reaching it

(Exact values depend on ν, α; the above is illustrative for $\nu = 10{,}000$, $\alpha = 0.05$.)

The sequential multiplier starts high and shrinks toward $z_{\alpha/2}$ as more data arrives. The gap — the "price of peeking" — decreases but never fully closes. For most practical experiments (thousands to hundreds of thousands of observations), the price is modest: 10–40% wider than a classical CI.

The role of ν and M

Recall from Act 8: ν controls *which sample size* the boundary is tightest for. Schmit & Miller set:

$$\nu = M \cdot \hat{\sigma}^2$$

where M is the **expected total sample size** (planned experiment duration $\times$ daily traffic).

If...	**Then...**
M is too small (underestimate traffic)	CI is tight early but wide late
M is too large (overestimate traffic)	CI is wide early but tight late
M matches actual traffic	CI is tightest around the planned end date

Setting M is like aiming a spotlight: you choose where the beam is brightest. Aim at the expected experiment end, and you get the tightest CI right when you need to make the decision. If the experiment runs longer or shorter than planned, the CI is still valid — just not optimally tight.

Reporting: from treatment effect to relative lift

Eppo reports the relative lift and its CI:

$$\text{Relative lift}(t) = \frac{\hat{\tau}(t)}{\bar{Y}^*_{\text{control}}(t)}, \qquad \text{CI}_{\text{lift}}(t) = \frac{\text{CI}(t)}{\bar{Y}^*_{\text{control}}(t)}$$

Dividing by the control mean converts the absolute difference into a percentage. "The treatment increased purchases by €3 per user" becomes "the treatment increased purchases by +3%."

Decision rule

What you see	Colour	What it means
CI entirely above 0	Green	Positive effect. Ship it.
CI entirely below 0	Red	Negative effect. Revert.
CI crosses 0, experiment ongoing	Grey	Inconclusive. Keep collecting.
CI crosses 0, experiment ended	Grey	No significant effect. Decide pragmatically.

At any of these moments, the decision is valid. This is the payoff of the entire mathematical journey from Act 0: Ville's inequality, applied through eight acts of increasingly practical machinery, guarantees that the false positive rate stays $\leq \alpha$ no matter when you look.

Key Takeaway

What Eppo actually uses (summary):

- **Framework:** Howard et al. (2021) confidence sequences (not the mSPRT of Johari et al. 2017).
- **Boundary:** The Normal mixture boundary $u(v) = \sqrt{(v+\nu)\log\frac{v+\nu}{\nu\alpha^2}}$.
- **Tuning:** $\nu = M\hat{\sigma}^2$, where M is the expected total sample size.
- **Variance:** Estimated from data (not assumed known).
- **Noise reduction:** Generalised CUPED via per-group regression adjustments.
- **Estimand:** Relative lift $\mu_1/\mu_0 - 1$.
- **Decision:** Stop when CI no longer crosses zero, valid at any time.

14 Act 13 — What If You Don't Have Eppo? (The Poor Man's Sequential Test)

Acts 9–12 described Eppo's sophisticated pipeline. But many teams run experiments without a dedicated platform. This act presents three progressively better approaches to peek-safe testing that anyone can implement with a spreadsheet or a few lines of code — and compares them head-to-head with Eppo's confidence intervals.

14.1 The Simulation

Simulation

What is being simulated: A single A/B experiment monitored at K pre-planned analysis times, showing four methods side-by-side.

Panel 1 — Four confidence bands: The same experiment, with four CI bands evolving over time:

- **Red (Bonferroni):** Widest. Simple but conservative.
- **Orange (Pocock):** Constant threshold, tighter than Bonferroni.
- **Blue (O'Brien–Fleming):** Starts very wide, narrows to near-classical at the final analysis.
- **Green (Eppo / Howard):** Continuous monitoring. No need to pre-specify K.

Vertical dashed lines mark the K planned analysis times. Between analyses, the group-sequential methods (Bonferroni, Pocock, OBF) have no valid CI — only Eppo's band is defined everywhere.

Panel 2 — 10,000 experiments: Run 10,000 experiments under the null ($\delta = 0$). Counters show the false positive rate for each method:

- Classical CI with K peeks: $\gg 5\%$.
- Bonferroni: $\leq 5\%$ (often well below).
- Pocock: $\leq 5\%$ (closer to 5%).
- O'Brien–Fleming: $\leq 5\%$.
- Eppo: $\leq 5\%$.

Panel 3 — Efficiency comparison: Under a true effect ($\delta > 0$), show average stopping time (number of observations to reach significance) for each method.

Controls:

- Number of planned peeks K (2 to 20).
- True effect size δ (0 to 10%).
- Significance level α.
- Fixed seed toggle (deterministic replay).
- "Run 1,000 experiments" mode (batch statistics).

Key takeaway: All four methods control the false positive rate. The price of simplicity is wider intervals (slower detection). Eppo's approach is the most efficient, but even the simplest method (Bonferroni) is far better than naive peeking.

14.2 Intuitive Explanation

Intuitive Explanation

You want to check your experiment every Monday. If you use a standard 95% CI each time, you'll get false positives far too often (Act 0).
The simplest fix: **be stricter at each check.** If you plan to peek 4 times, demand stronger evidence each time. The three methods below differ in *how* they distribute that strictness across the peeks.
Think of it as a budget. You have a total error budget of $\alpha = 5\%$.

- **Bonferroni** distributes the budget equally: α/K per peek.
- **Pocock** distributes it more cleverly (using the correlation between test statistics at different peeks), but still equally across peeks.
- **O'Brien–Fleming** front-loads the budget: almost no spending early, most of it saved for the final analysis.

Eppo's approach (Acts 8–12) doesn't need a fixed budget at all — it works for *any* number of peeks, continuously.

14.3 Method 1: Bonferroni Correction (Simplest)

The idea

Before the experiment starts, commit to checking exactly K times (e.g. every Monday for 4 weeks: $K = 4$). At each check, use a significance level of α/K instead of α.

The recipe

1. Choose K (number of planned peeks) and α (e.g. 0.05).
2. At each peek $k = 1, \ldots, K$, compute the standard CI using the adjusted significance level:
$$\text{CI}_k = \hat{\tau}(t_k) \ \pm \ \hat{\sigma}_{\hat{\tau}}(t_k) \cdot z_{\alpha/(2K)}$$
3. If $0 \notin \text{CI}_k$: stop and declare significance.
4. If $k = K$ and $0 \in \text{CI}_K$: no significant effect.

K (peeks)	α/K	$z_{\alpha/(2K)}$	CI multiplier vs. 1.96
1	0.050	1.96	1.00× (no correction)
2	0.025	2.24	1.14×
4	0.0125	2.50	1.28×
8	0.00625	2.73	1.39×
12	0.00417	2.87	1.46×
20	0.0025	3.02	1.54×
52	0.00096	3.29	1.68× (weekly for a year)

With 4 planned peeks, replace $z = 1.96$ with $z = 2.50$. Your CI is 28% wider at each peek, but you can safely check 4 times without inflating errors.

Pros and cons

+ **Dead simple.** One line of code: change `1.96` to `z_alpha/(2K)`.
+ **Always valid.** Works for any test statistic, any distribution.
− **Conservative.** The CI is wider than necessary because Bonferroni ignores the correlation between test statistics at successive peeks.
− **Scales poorly.** As K grows, the correction becomes increasingly harsh.

14.4 Method 2: Pocock Boundaries (1977)

The idea

Like Bonferroni, pre-specify K equally-spaced analysis times. But instead of Bonferroni's crude α/K split, use a constant critical value c_P that accounts for the correlation between test statistics at different peeks.

Because the test statistics $Z_1, Z_2, \ldots, Z_K$ at successive analyses share overlapping data, they are positively correlated. Pocock's method exploits this to derive a tighter threshold.

The recipe

1. Choose K and α.
2. Look up (or compute) the Pocock critical value $c_P(K, \alpha)$.
3. At each peek k, compute the z-statistic: $Z_k = \hat{\tau}(t_k)/\hat{\sigma}_{\hat{\tau}}(t_k)$.
4. If $|Z_k| > c_P$: stop and declare significance.

K	c_P (Pocock)	$z_{\alpha/(2K)}$ (Bonf.)	Improvement
2	2.18	2.24	3% tighter
4	2.36	2.50	6% tighter
8	2.51	2.73	8% tighter
12	2.60	2.87	9% tighter
20	2.69	3.02	11% tighter

Pocock gives a constant threshold that is a few percent lower than Bonferroni's. The saving is small when K is small (3% for 2 peeks) but grows as the number of peeks increases (11% for 20 peeks).
The key advantage: the threshold is the same at every peek, so it is almost as simple to use as Bonferroni.

Pros and cons

+ **Tighter than Bonferroni.**
+ **Constant threshold** — easy to remember and apply.
− **Requires equally-spaced analyses.**
− **Spends too much α early.** Because the same threshold is used early and late, it is harder to reject at the planned final analysis than a classical test.

14.5 Method 3: O'Brien–Fleming Boundaries (1979)

The idea

Instead of a constant threshold, use a threshold that *decreases* over time. At the first peek, demand very strong evidence; at the last peek, demand nearly the same evidence as a classical test.

The recipe

1. Choose K and α.
2. At peek k (of K total), use the critical value:

$$c_k^{\text{OBF}} \approx z_{\alpha/2} \cdot \sqrt{K/k}$$

   (This is the asymptotic approximation. Exact values are computed numerically.)
3. If $|Z_k| > c_k^{\text{OBF}}$: stop and declare significance.

Peek k (of K=4)	c_k^{OBF}	c_P (Pocock)	$z_{\alpha/(2K)}$ (Bonf.)	$z_{\alpha/2}$ (classical)
1 (25% of data)	4.05	2.36	2.50	1.96
2 (50% of data)	2.86	2.36	2.50	1.96
3 (75% of data)	2.34	2.36	2.50	1.96
4 (100% of data)	2.02	2.36	2.50	1.96

O'Brien–Fleming is extremely conservative early (threshold 4.05 at the first peek — you'd almost never stop that early) but barely penalises the final analysis (2.02 vs. 1.96 — only 3% wider).
This is the most efficient approach: if you run the experiment to completion, you pay almost nothing for the option to peek. The cost is that early stopping is unlikely unless the effect is very large.

Pros and cons

+ **Most efficient** of the three simple methods. Final-analysis CI is barely wider than classical.
+ **Catches large effects early** (if the effect is huge, even the high early threshold is exceeded).
− **Varying threshold** — slightly harder to implement and communicate.
− **Requires equally-spaced analyses.**

14.6 Head-to-Head Comparison

	Bonferroni	**Pocock**	**OBF**	**Eppo**
Pre-specify K?	Yes	Yes	Yes	No
Threshold type	Constant	Constant	Decreasing	Continuous
CI at peek 1 (K=4)	$2.50 \times \hat{\sigma}$	$2.36 \times \hat{\sigma}$	$4.05 \times \hat{\sigma}$	$\sim 3.8 \times \hat{\sigma}$
CI at final analysis (K=4)	$2.50 \times \hat{\sigma}$	$2.36 \times \hat{\sigma}$	$2.02 \times \hat{\sigma}$	$\sim 2.3 \times \hat{\sigma}$
Valid between peeks?	No	No	No	Yes
Variance reduction?	Manual	Manual	Manual	Built-in
Implementation effort	1 line	Lookup table	Formula	Platform

Key Takeaway

Which method should you use?

- **No platform, minimal effort:** Bonferroni. Commit to K peeks. Replace 1.96 with $z_{\alpha/(2K)}$. Done.
- **No platform, moderate effort:** O'Brien–Fleming. Nearly as efficient as Eppo at the final analysis, but requires pre-specifying K and cannot peek between scheduled analyses.
- **Need continuous monitoring or variance reduction:** Use a platform like Eppo. The machinery of Acts 8–12 is what you're paying for — continuous validity, regression adjustment, and no need to pre-commit to K.

The bottom line: even the crudest correction (Bonferroni) is *vastly* better than naive peeking. If you are checking results without any correction, you should start with Bonferroni *today*.

Historical Note

Stuart Pocock (1977) and **Peter O'Brien & Thomas Fleming** (1979) developed these group sequential methods for clinical trials, where stopping a trial early can save lives. Their work sits between Wald's SPRT (1945) and Howard et al.'s continuous framework (2021) in the historical timeline. The key insight of all three: if you want to peek, you must pay — but the price is negotiable.

14.7 Between Acts — Transition Cards

Each transition features a brief interstitial card with a bridge sentence and a "Key insight so far" box.

Transition	Bridge sentence
Act 0 → 1	"To build a peek-safe test, we first need the right building block. It starts with a coin flip."
Act 1 → 2	"Now let's put money on each step — and learn why no strategy can beat a fair game."
Act 2 → 3	"Fair games can't be beaten. Now let's build a detective — a tool for telling two coins apart."
Act 3 → 4	"We can measure evidence with a likelihood ratio. But here's the surprise: that evidence score is itself a martingale."
Act 4 → 5	"The likelihood ratio is a martingale. Now let's see why that single fact gives us the peeking guarantee."
Act 5 → 6	"We have the theory: peeking is safe. Now let's turn it into a practical decision procedure."
Act 6 → 7	"SPRT is optimal but inflexible. What if we don't know the effect size? Robbins had a radical idea."
Act 7 → 8	"The mixture approach works, but can we make it nonparametric? Howard et al. found the answer."
Act 8 → 9	"The theory is complete. Now let's see what problems a real A/B testing platform must solve."
Act 9 → 10	"Three problems, three solutions. Now let's walk through Eppo's pipeline, step by step."
Act 10 → 11	"The pipeline uses regression adjustment to remove noise. Let's see exactly how and why that works."
Act 11 → 12	"The noise is reduced. Now let's wrap everything in a sequential confidence interval and make the final decision."
Act 12 → 13	"Eppo solves everything — but what if you don't have Eppo? Three simple alternatives, from crude to clever."

14.8 Persistent Controls

A slider panel available throughout all acts:

- Probability p (or effect size δ from Act 3 onward)
- Significance level α
- Number of steps / observations n
- Simulation speed
- Number of simultaneous runs

14.9 Persistent Sidebar

A collapsible sidebar shows the act list and current position, allowing quick navigation to any act.

14.10 Visual Design Principles

- The same visual object (path / process) carries through all fourteen acts, relabelled at each stage.
- Threshold lines appear gradually: none in Acts 0–1; $\pm\sqrt{n}$ envelope in Act 1; both SPRT thresholds in Act 6; only the upper threshold in Act 7; confidence bands from Act 8 onward.
- Formula panels slide in/out; not shown by default on first reading.
- Every formula has a plain-language translation directly below it.
- Key terms on first use are bolded and defined in-line.

14.11 Colour Scheme

A consistent colour palette is used across all acts:

Element	Colour	Usage
Control group	Blue	Dots, means, CI bands for the control arm
Treatment group	Green	Dots, means, CI bands for the treatment arm
CI — positive effect	Green	CI entirely above 0: ship the feature
CI — negative effect	Red	CI entirely below 0: revert
CI — inconclusive	Grey	CI crosses 0: keep collecting
Classical CI (invalid)	Dashed grey	Shown for comparison; not valid under peeking
Raw outcomes (unadjusted)	Red	Acts 11–12: unadjusted CI / scatter
Adjusted outcomes	Blue	Acts 11–12: regression-adjusted CI / scatter

14.12 Simulation Interaction Standards

Every simulation across all acts must follow a consistent structure:

1. **What is being simulated** — one-sentence description.
2. **Controls** — sliders, toggles, and buttons.
3. **Visual outputs** — panels, charts, counters.
4. **Key takeaway** — what the user should learn.

All simulations must support:

- **Interactive controls:** sliders and toggles respond in real time.
- **Fixed seed toggle:** deterministic replay for reproducible exploration.
- **"Run 1,000 experiments" mode:** batch mode showing aggregate statistics (false positive rate, average stopping time, power).
- **"Reset simulation" button:** returns all parameters and visuals to their default state.

14.13 Tooltips

Hoverable tooltips appear on key quantities throughout all acts:

- **CI:** "A range of plausible values for the treatment effect, valid at every moment since the experiment started."
- **Lift ($\hat{\tau}(t)/\hat{\mu}_0(t)$):** "The percentage change in the metric caused by the treatment, relative to control."
- **Variance:** "How spread out individual outcomes are. Higher variance means more noise and slower experiments."
- **Sequential multiplier:** "The factor that replaces 1.96 in a classical CI to guarantee validity under continuous monitoring."

14.14 Transitions Between Acts

Acts transition with smooth scroll animations:

- Clicking "Next act" smoothly scrolls to the next section.
- The persistent sidebar highlights the current act.
- Simulation state from the previous act can optionally carry forward (e.g. the same dataset used in Acts 10–12).

15 Summary: The Through-Line

Act	Object	Represents	Key idea
0	Repeated p-values	The peeking problem	Checking repeatedly inflates FP
1	Random walk S_n	Position	SD grows as $\sqrt{n}$
2	Winnings M_n	Gambler's fortune	Martingale; Doob: $\mathbb{E}[M_\tau] = M_0$
3	Likelihood ratio Λ_n	Evidence against H_0	Comparing hypotheses flip by flip
4	Λ_n under H_0	LR is a martingale	The cancellation proof; Λ_0=1, $\Lambda_n \geq 0$
5	Markov + Ville	Anytime-valid guarantee	$\mathbb{P}(\Lambda_n \text{ ever} \geq 1/\alpha) \leq \alpha$
6	SPRT	Wald's decision rule	Optimal but requires known δ
7	Λ_n^H (mixture LR)	Robust evidence	Robbins' mixtures $\rightarrow$ Johari's mSPRT
8	Confidence sequence	Modern framework	Howard: nonparametric, estimated σ^2
9	Peeking vs. variance	Three problems	Peeking, high variance, heterogeneous users
10	Eppo pipeline	Step-by-step factory	Randomise $\rightarrow$ adjust $\rightarrow$ CI $\rightarrow$ decide
11	Regression adjustment	Noise removal	Generalised CUPED; $\text{Var}(Y^*) = \text{Var}(Y)(1 - R^2)$
12	Eppo's sequential CI	Final output	Sequential multiplier; $\nu = M\hat{\sigma}^2$; decide anytime
13	DIY alternatives	Simple approximations	Bonferroni, Pocock, OBF vs. Eppo

The core message: All fourteen acts build on the same mathematical structure — non-negative (super)martingales and Ville's inequality. The anytime-valid guarantee flows from a single elegant fact:

A fair game cannot be beaten by choosing when to quit.

The evolution across eight decades:

$$\underset{1939}{\text{Ville}} \rightarrow \underset{1945}{\text{Wald (SPRT)}} \rightarrow \underset{1970}{\text{Robbins (mixtures)}} \rightarrow \underset{2017}{\text{Johari (mSPRT)}} \rightarrow \underset{2021}{\text{Howard (CS)}} \rightarrow \underset{2024}{\text{Eppo}}$$

16 References

1. **Ville, J.** (1939). *Étude critique de la notion de collectif.* Gauthier-Villars, Paris. — Original proof of Ville's inequality for non-negative supermartingales. *[Used in Acts 5, 7, 8.]*

2. **Wald, A.** (1945). Sequential tests of statistical hypotheses. *Annals of Mathematical Statistics*, 16(2), 117–186. `https://doi.org/10.1214/aoms/1177731118` — Foundational 70-page monograph defining the SPRT, proving the fundamental inequalities for error probabilities, deriving the ASN formulas, and showing 47–63% savings over fixed-sample tests. *[Used in Act 6.]*

3. **Wald, A. & Wolfowitz, J.** (1948). Optimum character of the sequential probability ratio test. *Annals of Mathematical Statistics*, 19(3), 326–339. `https://doi.org/10.1214/aoms/1177730197` — Proof that the SPRT is exactly optimal: among all sequential tests with the same error probabilities, it minimises expected sample size under both H_0 and H_1. *[Used in Act 6.]*

4. **Doob, J. L.** (1953). *Stochastic Processes.* Wiley, New York. — Definitive treatment of martingale theory including the Optional Stopping Theorem. *[Used in Act 2.]*

5. **Robbins, H.** (1970). Statistical methods related to the law of the iterated logarithm. *Annals of Mathematical Statistics*, 41(5), 1397–1409. `https://doi.org/10.1214/aoms/1177696786` — Introduced the mixture likelihood ratio approach and confidence sequences. Showed that weighting over alternatives via a mixing distribution produces a martingale amenable to Ville's inequality. Derived closed-form boundaries for the Normal case (including the root of the Normal mixture that Howard et al. later generalised) and connected boundary growth rates to the law of the iterated logarithm. *[Used in Act 7.]*

6. **Deng, A., Xu, Y., Kohavi, R., & Walker, T.** (2013). Improving the sensitivity of online controlled experiments by utilizing pre-experiment data. *Proc. WSDM*, 123–132. `https://doi.org/10.1145/2433396.2433413` — The original CUPED paper. Optimal adjustment coefficient is the OLS slope $\theta^* = \mathrm{Cov}(Y, X)/\mathrm{Var}(X)$; variance is reduced by a factor of $1-\rho^2$. Practical results at Microsoft Bing: 45–52% variance reduction with 2-week pre-period. *[Used in Acts 9–11.]*

7. **Johari, R., Pekelis, L., & Walsh, D.J.** (2017). Always valid inference: Continuous monitoring of A/B tests. *Operations Research*, 65(6), 1651–1667. `https://doi.org/10.1287/opre.2017.1607` — Applied the mixture SPRT to A/B testing. Derived the closed-form mSPRT for Normal observations, always-valid p-values and confidence intervals, and guidance for calibrating τ. Assumes known variance. *[Used in Act 7.]*

8. **Howard, S. R., Ramdas, A., McAuliffe, J., & Sekhon, J.** (2021). Time-uniform, nonparametric, nonasymptotic confidence sequences. *Annals of Statistics*, 49(2), 1055–1080. `https://doi.org/10.1214/20-AOS1991` — Developed the sub-ψ framework for confidence sequences. Key result: the Normal mixture boundary $u(v) = \sqrt{(v+\nu)\log\frac{v+\nu}{\nu\alpha^2}}$. Nonparametric (works beyond Normal data), nonasymptotic (finite-sample guarantees), and allows estimated variance. **This is the frame-**

work Eppo adopted. *[Used in Acts 8, 10, 12.]*

9. **Schmit, M. & Miller, T.** (2024). Sequential confidence intervals for comparing two proportions with variance reduction. *Eppo Technical Report.* — Eppo's implementation. Combines Howard et al.'s Normal mixture confidence sequence with regression-adjusted variance reduction (generalised CUPED). Estimates variance (does not assume known σ^2). Tuning parameter is ν (set from expected sample size M). Reports relative lift with sequential confidence intervals. *[Used in Acts 9–12.]*

Appendix: What Changed from v3

1. **Major correction:** Eppo does *not* use the mSPRT (Johari et al. 2017). They use the confidence sequence framework of Howard et al. (2021), specifically the Normal mixture boundary. Acts 8 and 9 completely rewritten.
2. **GAVI removed.** The term "GAVI" does not appear in the Schmit & Miller paper. The tuning parameter is ν, set from the expected sample size M, not from an empirical distribution of past effect sizes.
3. **Robbins (1970) properly credited.** His mixture approach and confidence sequences are now covered in Act 7A, before Johari's mSPRT (Act 7B). This corrects the historical attribution.
4. **Act 6 enriched:** Added Wald's ASN formula, savings table (47–63%), the random walk view, truncation, and the optimality conjecture resolved by Wald & Wolfowitz (1948).
5. **CUPED corrected:** Added the validity condition ($\mathbb{E}[X^{(t)}] = \mathbb{E}[X^{(c)}]$), the warning against post-treatment covariates (from Deng et al. 2013), recommended pre-period length, practical results from Bing.
6. **Notation cheat sheet removed:** All notation is now explained inline at first use.
7. **Act numbering:** Now 14 acts (0–13) instead of 10 (0–9). Former Act 9 (Eppo, all-in-one) expanded into Acts 9–12 following detailed guidance: Act 9 (problem Eppo solves), Act 10 (pipeline step-by-step), Act 11 (variance reduction via regression), Act 12 (sequential CIs in Eppo). Act 13 added: DIY alternatives (Bonferroni, Pocock, O'Brien–Fleming) with head-to-head Eppo comparison.
8. **References:** Each reference now notes which acts use it and describes the specific contribution more precisely.
9. **Historical notes** added throughout (Ville's 1939 thesis, Wald's 1945 optimality conjecture, Wald–Wolfowitz 1948 resolution).
10. **Through-line updated:** Timeline from Ville (1939) through Eppo (2024) added to the summary.